

IoT Cloud Integration & Real-Time Remote Monitoring

ABSTRACT

This project presents the design and implementation of a cloud-connected IoT monitoring system using an ESP32 microcontroller simulated on the Wokwi online platform. Building upon the Smart Automation System developed in Task-3, this task extends the system by integrating WiFi-based internet connectivity to transmit live sensor data — including temperature, humidity, and light intensity — to a free cloud IoT platform (ThingSpeak/Blynk) in real time. The ESP32 reads sensor values at regular intervals and uploads them to the cloud via HTTP API calls, where the data is visualized on an interactive web dashboard featuring live graphs and real-time readings accessible from any device and location. This implementation demonstrates the complete IoT data pipeline from physical sensing to cloud visualization, providing practical exposure to embedded systems networking, cloud platform integration, and real-time dashboard creation — skills that are fundamental to modern Smart Home and Industry 4.0 applications.

OBJECTIVE

The objective of Task-4 is to upgrade the Smart Automation System built in Task-3 into a fully cloud-connected IoT device capable of transmitting live sensor data over the internet and enabling real-time remote monitoring from any location through a web or mobile dashboard.

In this task, an ESP32 microcontroller (simulated using the Wokwi online simulator) is programmed to read sensor values such as temperature, humidity, light intensity, or motion, and continuously upload this data to a free cloud IoT platform such as ThingSpeak or Blynk using WiFi and HTTP/API communication.

Tools & Platform Used

Item	Details
Microcontroller	ESP32 (Wokwi Simulator)
Sensors Used	LDR Sensor
IoT Cloud Platform	ThingSpeak

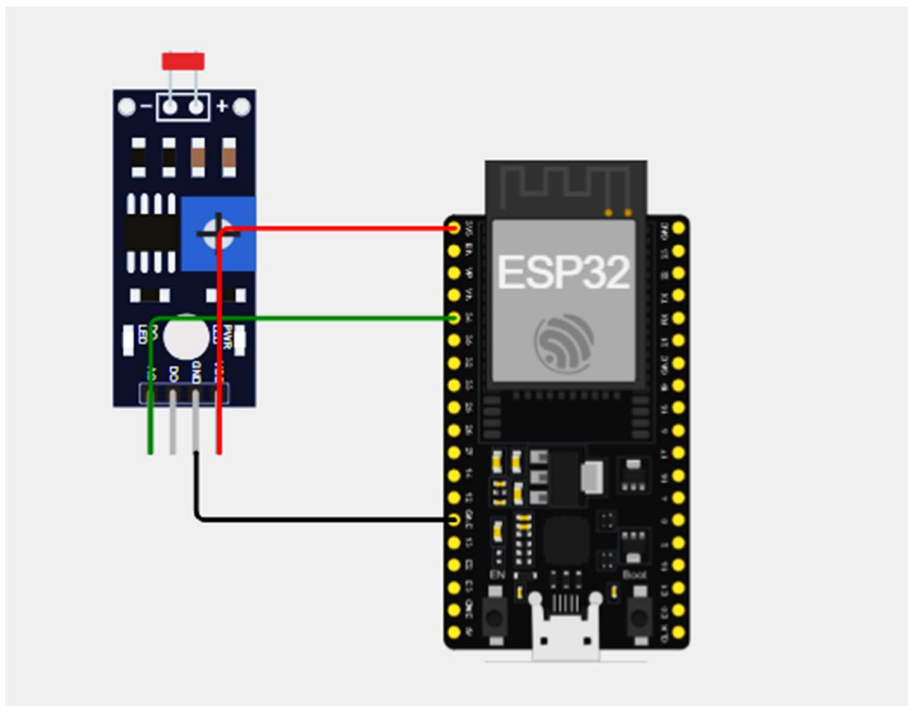
Circuit Description :

The ESP32 is connected with the LDR sensor module to measure light intensity and upload the readings to the ThingSpeak cloud platform through WiFi communication.

Connection Table

LDR Sensor	Pin ESP32
VCC	VIN
GND	GND
AO	GPIO 34

The VCC pin of the LDR module is connected to the VIN pin of the ESP32 to provide power supply to the sensor module. The GND pin of the sensor is connected to the GND pin of the ESP32 to establish a common ground connection. The AO (Analog Output) pin of the LDR module is connected to GPIO 34 of the ESP32, which is an analog input pin used to read sensor values.



WORKING PRINCIPLE

The system works by continuously monitoring the surrounding light intensity using an LDR sensor and sending the collected data to the ThingSpeak cloud platform through the ESP32 WiFi module. The uploaded data is then displayed on a real-time dashboard in graphical form for remote monitoring.

Initially, the LDR (Light Dependent Resistor) senses the amount of light falling on its surface. The resistance of the LDR changes according to the light intensity. When the light intensity increases, the resistance of the LDR decreases, and when the light intensity decreases, the resistance increases. This variation is converted into an analog voltage signal by the sensor module.

The analog output pin (AO) of the LDR module is connected to GPIO 34 of the ESP32. The ESP32 contains an inbuilt Analog-to-Digital Converter (ADC) which converts the analog voltage into a digital value. The sensor readings are obtained using the `analogRead()` function. The ADC value lies in the range:

$$0 \leq \text{ADC Reading} \leq 4095$$

Higher values indicate higher light intensity, while lower values indicate lower light intensity.

After reading the sensor value, the ESP32 connects to the WiFi network using the SSID and password provided in the program code. Since the ESP32 has a built-in WiFi module, no external WiFi hardware is required. Once the WiFi connection is established successfully, the ESP32 becomes capable of communicating with internet-based cloud platforms.

The ESP32 then creates an HTTP GET request containing the ThingSpeak Write API key and the sensor reading. The data is sent to the ThingSpeak server through the internet using the API structure .

ThingSpeak receives the uploaded sensor values and stores them in the cloud database. The platform automatically generates a live graph showing the variation of light intensity with respect to time. As new sensor readings are uploaded every 15 seconds, the dashboard updates continuously, enabling real-time remote monitoring of environmental light conditions from anywhere using the internet.

Thus, the project demonstrates an IoT-based cloud monitoring system where sensor data is acquired, processed, transmitted over WiFi, and visualized remotely through a cloud dashboard.

SOURCE CODE

```
1  #include <WiFi.h>
2  #include <HTTPClient.h>
3
4  // WiFi (Nokwi)
5  const char* ssid = "Nokwi-GUEST";
6  const char* password = "";
7
8  // ThingSpeak
9  String apiKey = "0V59AQ2KP0M005ML";
10 const char* server = "http://api.thingspeak.com/update";
11
12 int ldrPin = 34; // A0 connected here
13
14 void setup() {
15     Serial.begin(115200);
16
17     WiFi.begin(ssid, password);
18     Serial.print("Connecting to WiFi");
19
20     while (WiFi.status() != WL_CONNECTED) {
21         delay(500);
22         Serial.print(".");
23     }
24
25     Serial.println("\nConnected!");
26 }
27
28 void loop() {
29     int ldrValue = analogRead(ldrPin); // Read analog value (0-4095)
30
31     Serial.print("Light Value: ");
32     Serial.println(ldrValue);
33
34     if (WiFi.status() == WL_CONNECTED) {
35         HTTPClient http;
36         String url = server + String("?api_key=") + apiKey + "&field1=" + ldrValue;
37         http.begin(url);
38         int httpResponseCode = http.GET();
39         Serial.print("Response: ");
40         Serial.println(httpResponseCode);
41         http.end();
42     }
43
44     delay(15000); // ThingSpeak limit
45 }
```

OUTPUT / RESULTS

After powering the ESP32 and uploading the program, the board first connects to the WiFi network. Once the connection is established successfully, the ESP32 starts reading the light intensity values from the LDR sensor connected to GPIO 34.

The sensor values displayed on the serial monitor continuously change according to the surrounding light conditions. When more light falls on the LDR sensor, the analog value changes, and when the light intensity decreases, the value changes accordingly. These values are generated by the Analog-to-Digital Converter (ADC) of the ESP32 and are displayed in the serial monitor for observation and debugging purposes.

```
Light Value: 3466
Response: 200
Light Value: 2653
Response: 200
Light Value: 1750
Response: 200
Light Value: 1049
Response: 200
Light Value: 2045
Response: 200
Light Value: 3017
Response: 200
Light Value: 3906
Response: 200
Light Value: 3779
Response: 200
Light Value: 2340
Response: 200
Light Value: 1207
Response: 200
Light Value: 593
Response: 200
```

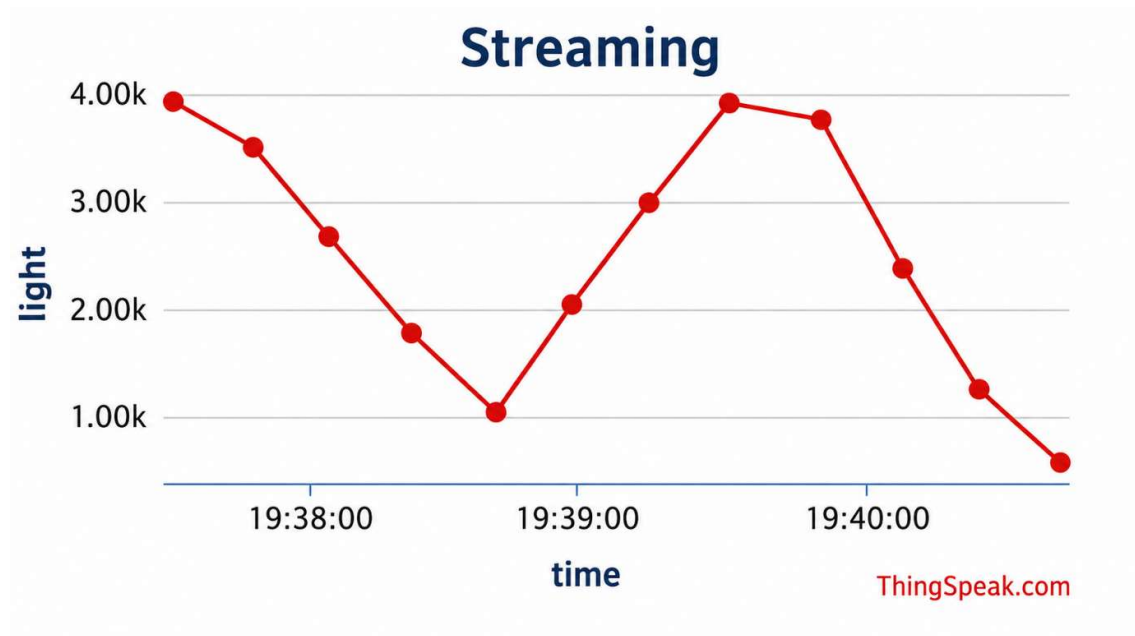
The “Light Value” represents the intensity of light detected by the sensor. Higher values indicate higher light intensity, while lower values indicate lower light intensity.

The sensor data is uploaded repeatedly after a fixed delay of 15 seconds because ThingSpeak allows updates after a minimum interval. Continuous successful responses indicate stable WiFi and cloud connectivity.

ThingSpeak Dashboard Output :

The ThingSpeak dashboard is used to visualize the uploaded sensor data in graphical form. After the ESP32 sends the light intensity values to the cloud server, ThingSpeak stores the received data and automatically updates the graph in real time.

The graph displays the variation of light intensity with respect to time. The horizontal axis represents time, while the vertical axis represents the sensor readings obtained from the LDR module. As the surrounding light changes, the graph continuously updates with new data points uploaded by the ESP32.



The dashboard can be accessed remotely from any device connected to the internet, making the system suitable for smart monitoring applications such as smart homes, industrial automation, and environmental monitoring systems.

CONCLUSION

The IoT-based light monitoring system using ESP32 and ThingSpeak was successfully designed and implemented. The project demonstrated how an embedded system can be connected to a cloud platform for real-time remote monitoring. The LDR sensor was able to detect changes in surrounding light intensity and generate corresponding analog voltage signals. These analog signals were read by the ESP32 microcontroller through its built-in ADC pins.

The ESP32 successfully processed the sensor readings and established a stable WiFi connection using its inbuilt wireless communication module. After connecting to the internet, the ESP32 continuously transmitted the sensor data to the ThingSpeak cloud server using HTTP API requests. The successful HTTP response codes displayed in the serial monitor confirmed proper communication between the hardware system and the cloud platform.

The ThingSpeak platform effectively stored the uploaded sensor data and displayed it in graphical form through a real-time dashboard. The graph continuously updated according to the changing light conditions, which verified the successful implementation of live cloud-based monitoring. The dashboard provided easy visualization and remote accessibility of sensor readings from any internet-connected device.

This project helped in understanding important concepts of Internet of Things (IoT), including sensor interfacing, analog data acquisition, WiFi communication, cloud integration, API-based data transmission, and real-time data visualization. The system also demonstrated how IoT technology can be used to monitor environmental conditions remotely and efficiently.

Overall, the project successfully achieved the objective of creating a cloud-connected IoT monitoring system using ESP32 and ThingSpeak. The developed system is reliable, low-cost, easy to implement, and can be further expanded for advanced smart monitoring and automation applications .